

7. Breaking a Simple Substitution Cipher

```
#include <stdio.h>
```

```
#include <string.h>
```

```
int main()
```

```
{
```

```
    char cipher[100];
```

```
    char plain[100];
```

```
    int i;
```

```
    printf("Enter ciphertext: ");
```

```
    scanf("%s", cipher);
```

```
    for(i = 0; cipher[i] != '\0'; i++)
```

```
    {
```

```
        switch(cipher[i])
```

```
        {
```

```
            case 'S': plain[i] = 'H'; break;
```

```
            case 'V': plain[i] = 'E'; break;
```

```
            case 'O': plain[i] = 'L'; break;
```

```
            case 'L': plain[i] = 'O'; break;
```

```
            default: plain[i] = cipher[i];
```

```
        }
```

```
    }
```

```
    plain[i] = '\0';
```

```
printf("Decrypted text: %s", plain);

return 0;
}
```

